

Prep Test 2: Functions and Polynomial Graphs

75 minutes | 100 points | 8 questions

Name: _____

Date: _____

Directions

- Show your work.
- Use interval notation where requested.
- Questions 6 and 8 require qualitative polynomial sketches. Exact extrema and a uniform vertical scale are not required.
- Suggested question times total 70 minutes, with a final 5 minutes for review.

Use the space provided; additional work may go on the back.

Suggested timing

Question	Points	Minutes
1. Graph a piecewise function	12	8
2. Transform a given graph	12	8
3. Graph both composition orders	18	12
4. Evaluate and analyze two graphs	12	8
5. Build a graph from its characteristics	10	7
6. Construct and sketch a polynomial	14	10
7. Compose functions and find a domain	12	9
8. Analyze and sketch a quartic	10	8

Final review: 5 minutes. Total: 75 minutes.

Question 1

8 minutes | 12 points

1 Graph a piecewise function

12 points

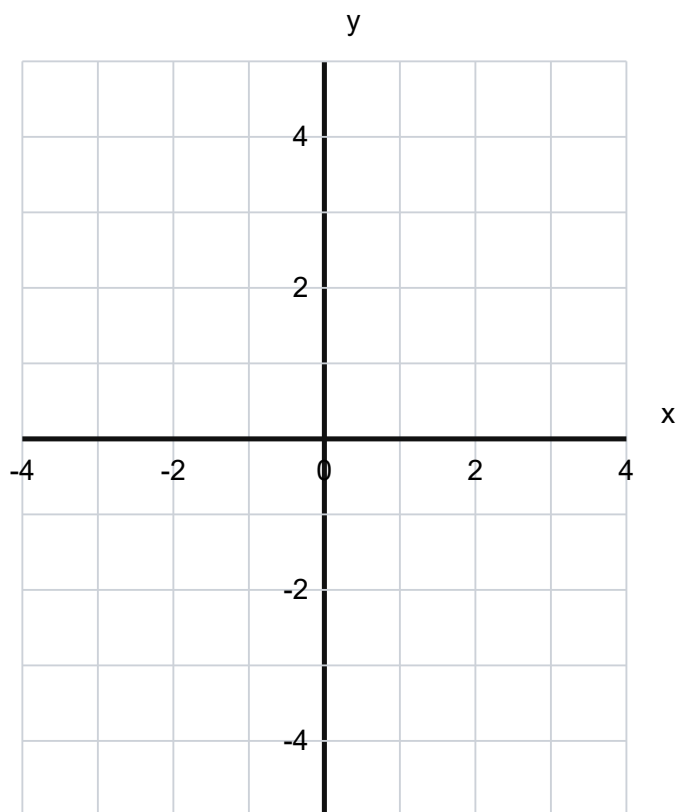
Let $p(x)=2x+4$ for $x<-1$, and $p(x)=3$ for $x\geq-1$.

(a) Find $p(-2)$, $p(-1)$, and $p(2)$. Show which branch you use.

(b) Graph p on axes $x=-4$ to 4 , $y=-5$ to 5 , with one unit per square. Continue each branch with an arrow. Label the boundary coordinates and show open or closed circles.

(c) Give the domain and range in interval or set notation.

Your graph: one unit per square



Remaining answers and reasoning

Question 2

8 minutes | 12 points

2 Transform a given graph

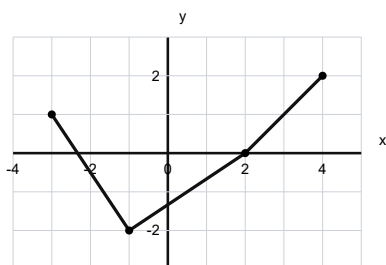
12 points

The graph of f is shown below. Its vertices are $(-3,1)$, $(-1,-2)$, $(2,0)$, and $(4,2)$, joined by straight segments, with included endpoints. Let $h(x) = -f(x - 2) + 1$.

(a) Graph h on the provided grid. Label all vertices with their coordinates and mark the endpoints. Use axes $x=-2$ to 8 , $y=-2$ to 4 , one unit per square.

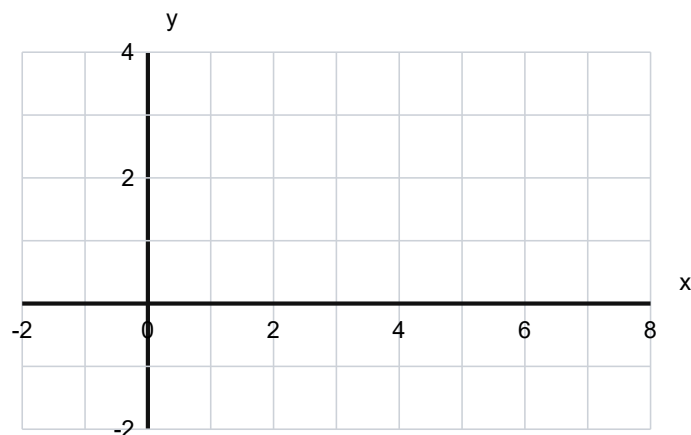
(b) Give the domain and range of h .

Given graph: f



Work space

Your graph: one unit per square



Remaining answers and reasoning

Question 3

12 minutes | 18 points

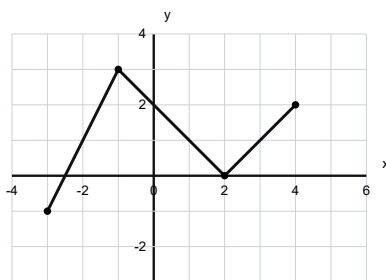
3 Graph both composition orders

18 points

The graph of f consists of straight segments through $(-3,-1)$, $(-1,3)$, $(2,0)$, $(4,2)$, in that order, with included endpoints. Let $g(x)=2-x$.

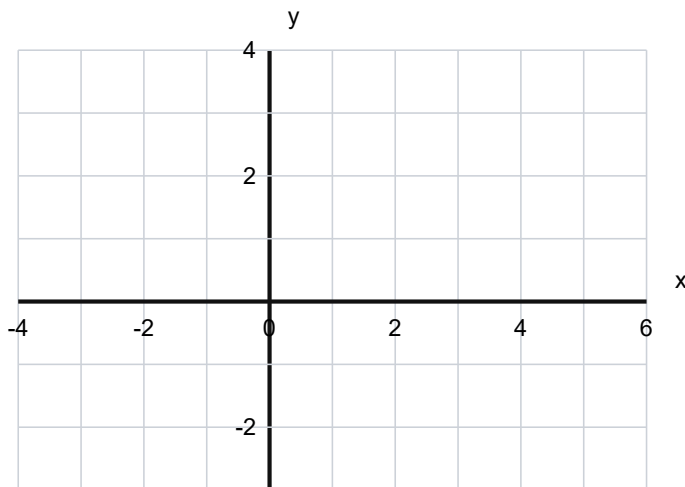
- Make a transformed-vertex table for $f(g(x))$ and for $g(f(x))$. Show the coordinate rule for each.
- On one grid, graph $f(g(x))$ with a solid line and $g(f(x))$ with a dashed line. Label every vertex with its coordinates and name each graph. Use axes $x=-4$ to 6 , $y=-3$ to 4 , one unit per square. Do not extend beyond the mapped endpoints.
- Give both domains. Explain why the source vertices appear in reversed left-to-right order in one composition.

Given graph: f



Work space

Your graph: one unit per square



Remaining answers and reasoning

Question 4

8 minutes | 12 points

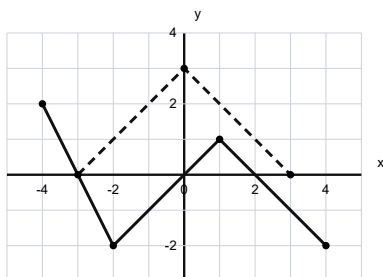
4 Evaluate and analyze two graphs

12 points

Both graphs consist of straight segments joining the listed vertices, with included endpoints. f : $(-4,2)$, $(-2,-2)$, $(1,1)$, $(4,-2)$. g : $(-3,0)$, $(0,3)$, $(3,0)$. The graphs below use axes $x=-5$ to 5 , $y=-3$ to 4 , one unit per grid line.

- Find $f(-1)$ and $g(-1)$.
- Find $g(f(-1))$ and $f(g(-1))$. Show your work.
- Give all zeros of f and its increasing and decreasing intervals.

Given graph: f solid; g dashed



Work space

Remaining answers and reasoning

Question 5

7 minutes | 10 points

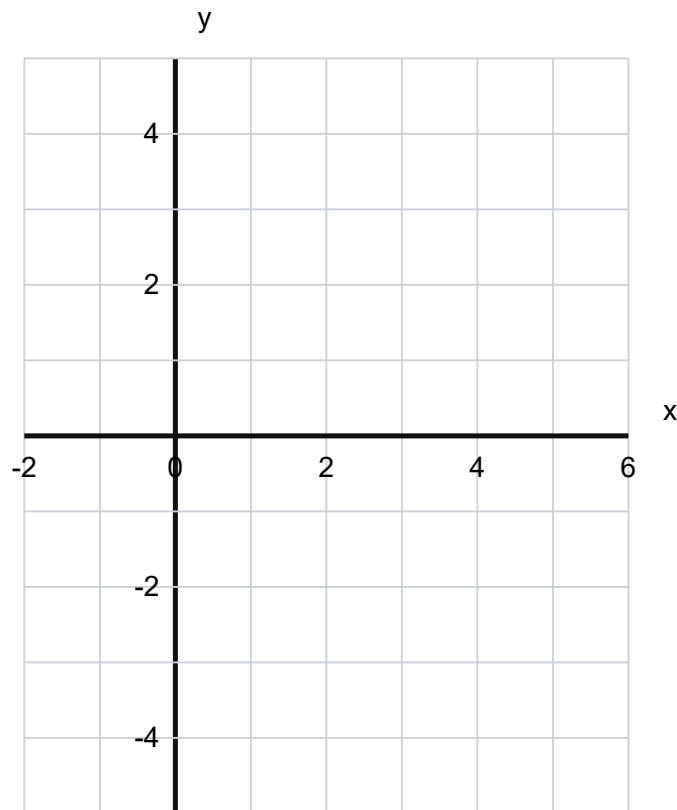
5 Build a graph from its characteristics

10 points

Find a function of the form $q(x) = a|x - h| + k$ with domain all real numbers, range $(-\infty, 4]$, maximum $(2, 4)$, and y-intercept $(0, 0)$.

- Find a , h , and k and write $q(x)$. Show how the intercept determines a .
- Graph q on axes $x=-2$ to 6 , $y=-5$ to 5 , one unit per square. Label the vertex and both x-intercepts; draw arrows.
- State where q increases and decreases, and confirm its domain and range.

Your graph: one unit per square



Remaining answers and reasoning

Question 6

10 minutes | 14 points

6

Construct and sketch a polynomial

14 points

A degree-5 polynomial P has exactly two distinct real zeros: -2 with multiplicity 2, and 1 with multiplicity 3. It passes through $(0, -8)$.

- (a) Find P in factored form and show how you determine its leading factor.
- (b) Make a qualitative sketch. Label both zeros and the y -intercept, show the correct ends, and show the graph's behavior at each zero. Exact extrema are not required; the sketch need not use a uniform vertical scale.
- (c) State the intervals where $P(x) > 0$ and $P(x) < 0$. Explain where the sign changes and where it does not.

Your qualitative sketch (label axes and intercepts)

Remaining answers and reasoning

Question 7

9 minutes | 12 points

7

Compose functions and find a domain

12 points

Let $f(t) = \sqrt{t}$, $g(x) = 9 - x^2$, and $H(x) = \frac{f(g(x))}{g(x)-5}$.

- (a) Write H as one expression in x.
- (b) Find the domain in interval notation. Show why each excluded value is excluded.
- (c) Is x=3 allowed? Is x=2 allowed? Explain using the original expression.

Remaining answers and reasoning

Question 8

8 minutes | 10 points

8

Analyze and sketch a quartic

10 points

Let $R(x) = -(x + 2)(x - 1)^2(x - 3)$.

- (a) State the degree, leading coefficient, zeros, and multiplicities. Find the y-intercept.
- (b) Make a qualitative sketch with all intercepts labeled and arrows on both ends. Show which zeros cross and which touch. Exact extrema and a uniform vertical scale are not required.
- (c) State the intervals where $R(x) > 0$ and $R(x) < 0$. Compare its end behavior with the polynomial in Question 6.

Your qualitative sketch (label axes and intercepts)

Remaining answers and reasoning
